Technical Appendix

Substitution and Complementarity: From the Firm Hierarchy

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1 Scope

This appendix gives the construction and validation of the measure of direction at the level of detail needed to evaluate it. Section 2 names the models and the data behind each step and summarizes the taxonomy and the exposure measure on which the ratings rest. Sections 3 to 5 give the rating design, the rated grid, the ratings and their conversion into substitution and complementarity contents, the validation of the measure, and its design choices with further results. The calibration of the silicon panel against a Delphi survey of domain experts is reported in Rony and Ciarli (2026). The prompts, runners and scoring code, the aggregated tables of the calibration, and the data, verdicts and scripts of the measure are in the repository that accompanies the proposal. The Delphi survey's respondent-level data, the analysis scripts of the calibration and the licensed patent extract are not.

2 Models and data

Table 1 reports, for each language-model step of the measure, the model used, its mode and date of access, and the information recorded. The two reviews were conducted with a proprietary model and the page judgments with a preview release, so that a repetition of these steps would reproduce their verdicts only approximately. The verdicts are therefore released in full, and all results from the released exposure scores onward can be reproduced exactly. The direction ratings were obtained from the model and sampling configuration selected in the calibration, an open-weight model accessed through an interface that routes each call to one of several hosting providers, and the provider was not recorded.

The patents come from one extract of patent applications with their DOCDB family, filing year, filing office, forward citations, CPC codes, title and abstract, in which each application is classified as digital or green. The taxonomy-matching corpus holds about 1.9 million digital patents. The digital exposure corpus holds some 880,000 patents after excluding applications filed only at the Chinese, Taiwanese or Russian office whose CPC codes no other office assigned, and the green exposure corpus about 5.7 million patents carrying ENV-TECH indicators. The extract is licensed and is not redistributed. Occupations are 427 of the 436 four-digit unit groups of ISCO-08 as implemented in the ESCO classification, which carries no occupations for the three armed-forces groups and six residual groups not elsewhere classified, represented by some 2,800 task sentences.

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Table 1: Language-model steps of the application.

Step	Model and access	Settings and record
Page judge of the taxonomy	Gemini 3 Flash, preview release (Google), through OpenRouter, early 2026	Temperature 0, top-p 1, pages in batches, prompt in the taxonomy pipeline code
Second review round of the taxonomy	Claude Opus 5.5 (Anthropic), through Claude Code, October 2026	Proposals adopted by the authors after inspection, every change recorded in the ledger
Link review of the exposure measure	Claude Opus 5.5 (Anthropic), through Claude Code, October 2026	Verdicts on named patents and links, recorded with the build
Direction ratings	DeepSeek-V3.1, open weights, through OpenRouter, September and October 2026	Temperature 0.3, top-p 0.3, five draws, JSON schema, prompt in Section 5.1

Industries are the 272 three-digit groups of NACE Rev 2, represented by some 850 activity sentences from the Eurostat NACE Rev 2 structure and its explanatory notes. The ten published exposure indices against which the exposure measure is checked are keyed to the US Standard Occupational Classification where they are not keyed to ISCO, and are mapped to ISCO-08 through a crosswalk between SOC-2010 and ISCO-08 derived from the Bureau of Labor Statistics, averaging over the SOC codes that map to one ISCO code.

The digital technologies are the 184 base technologies of a taxonomy built from Wikipedia and patents, 10 root domains and 43 intermediate clusters on 6,928 pages, and the green technologies are 66 nodes taken from the OECD ENV-TECH classification (Haščič and Migotto, 2015). Exposure is the citation-weighted semantic similarity between patent text and the task text of occupations and the activity text of industries, computed with the same text, embeddings and aggregation in both domains. Every verdict of the page judge, the taxonomy review and the link review is in the repository, with the code of both builds.

3 The substitution and complementarity measure

The direction measure rests on two constructs, substitution and complementarity, rated by a large language model under workplace personas, roles that the model is asked to adopt, for the technology-occupation-industry cells in which exposure concentrates (Section 5).

3.1 Unit and rating design

The rated unit is a triplet consisting of a technology node (a digital base technology or a green ENV-TECH node), a three-digit ISCO occupation group, and a NACE Rev 2 section, the finest occupation-industry resolution at which European employment data record the pair. The base-technology level is chosen because we expect specific technology labels to draw specific assessments, and ratings can be aggregated to coarser levels while the converse is not possible.

Two constructs are rated. Substitution is defined as the technology taking over the work itself, so that fewer workers are needed, and complementarity as the technology supporting or enhancing the workers, who continue to perform the work at higher quality or speed. Each construct is presented as an agreement statement and rated on a scale from 1 (strongly disagree) to 5 (fully agree), with

0 defined as do not know. The statements name no time horizon, because the rated technologies already exist, with patents filed within the exposure window 2008 to 2023. Section 5.1 gives the prompts verbatim. The prompt has no separate not-applicable option, and Section 5.5 shows that the zero is used as one. Each statement names the rated object as a digital or a green technology according to the domain of its node, and this domain label supplies the context in which the node's name is read.

Each rating is given under one of three personas, specified by a single role sentence. They are a manager who runs the team performing the work, an experienced worker who performs it, and a chief executive responsible for the organization's strategy and investment decisions. Ratings are stored per persona. The chief executive represents the side of the employment relation that decides on adoption, since investment and staffing decisions are the channel through which an assessment of substitution reaches labor demand.

The respondent model is DeepSeek-V3.1, a model with open weights, queried as `deepseek/deepseek-chat-v3.1` through the OpenRouter interface in September and October 2026, at temperature 0.3 and top-p 0.3, settings that limit the randomness of the sampled answers, as selected in our companion work on language-model survey panels (Rony and Ciarli, 2026). The interface routes each call to one of several hosting providers, and the provider was not recorded. Responses are constrained to a fixed machine-readable format, a JSON object holding one integer rating followed by a short justification. Each triplet receives 30 ratings, across two constructs, three personas and five independent draws. Each call, a single query to the model, carries one construct and one persona. In a test on an earlier version of the prompt, merging constructs or personas into one call moved the ratings away from those of separate calls, reduced the use of the zero category and raised the draw-to-draw noise (Section 5.2). Scale, response format and the one-construct, one-persona call follow Rony and Ciarli (2026).

3.2 The rated grid

The grid, the set of triplets put to the respondent model, is restricted to cells in which both exposure and employment indicate that the question is meaningful. It uses the exposure measure of Section 2, computed over the window 2008 to 2023 and aggregated from four-digit occupations to three-digit groups and from three-digit industries to sections by unweighted means over the exposed codes. A triplet qualifies if the exposure of its occupation group to the node and the exposure of its section to the node both exceed a threshold. The thresholds are common to all technologies, with one for occupations and one for industries in each domain. A node-specific threshold would admit the upper tail of nodes whose exposure is low throughout. Each threshold is the 70th percentile of the corresponding distribution of exposed cells (Section 5.3 gives the values). The grid was selected on the final taxonomy on 7 October 2026, with the window then in use, 2008 to 2023, before the window of the paper was fixed, and it was not re-selected on the paper's window. Triplets already in the earlier grid kept their September ratings, since their prompts were unchanged, and the new triplets were rated on 7 and 8 October. The descriptive statistics use exposure over 2009 to 2023, and the employment test builds the contents from the patents filed in the years of each outcome period.

The employment filter retains occupation-industry pairs accounting for at least 0.1 percent of the occupation's pooled 2011–2023 employment in the LFS, a floor that removes pairs with negligible employment. The grid has some 12,000 triplets, a third digital and two thirds green, on 112 digital and 62 green technologies. Entities enter the prompt under their names and green nodes under their ENV-TECH descriptions, since a bare classification code does not tell the respondent what the technology is.

3.3 The ratings

The runs yield 30 usable ratings for every triplet, some 360,000 in all. The few calls that failed or returned no rating were rerun. Table 2 and Figure 1 report their distribution.

Table 2: Ratings by domain and construct. The scale runs from 0 to 5, and the zero category is counted as its bottom.

Domain	Construct	Mean	SD	n
Digital	Complementarity	3.472	1.178	60,720
Digital	Substitution	2.223	0.869	60,720
Green	Complementarity	2.965	1.357	118,935
Green	Substitution	1.934	0.911	118,935

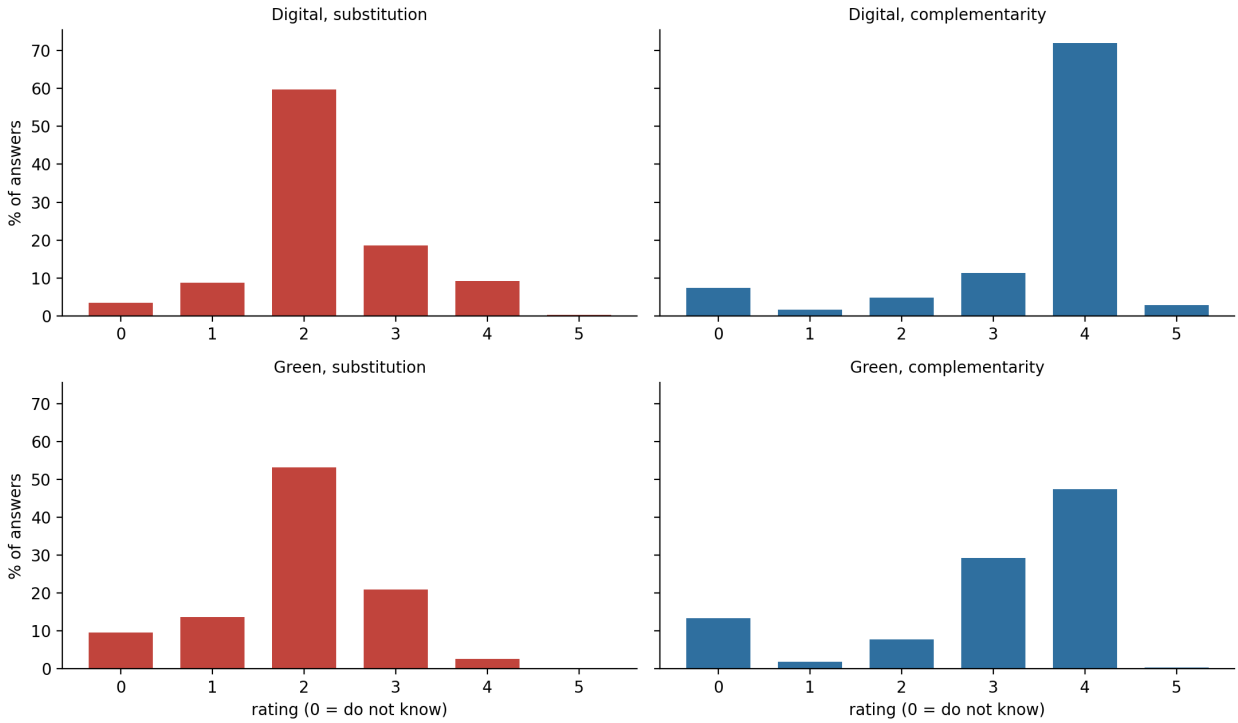


Figure 1: Distribution of the ratings by domain and construct. Substitution is modal at 2 and complementarity at 4 in both domains, the zero category is used most for green complementarity, and the top of the scale draws at most 3 percent of the ratings of any domain and construct.

Complementarity is rated above substitution in both domains. The respondent model rates substitution about a quarter of a scale point lower than experts do (Rony and Ciarli, 2026), so the level of this gap is partly a property of the model. The analysis below relies on its variation across technologies, occupations and industries. Mean complementarity exceeds mean substitution by more than one scale point in both domains, and at the triplet level, averaging the fifteen ratings per construct, substitution is rated above complementarity in about one triplet in ten. The two constructs move together across triplets, with a Pearson correlation of 0.69 in the digital and 0.83 in the green domain (Figure 2). Both rise where the technology affects the occupation's work, and

direction is the gap between them. Section 3.4 keeps both as levels. The personas move the level of the ratings and leave their ranking, and the zero category functions as a no-effect code (Section 5).

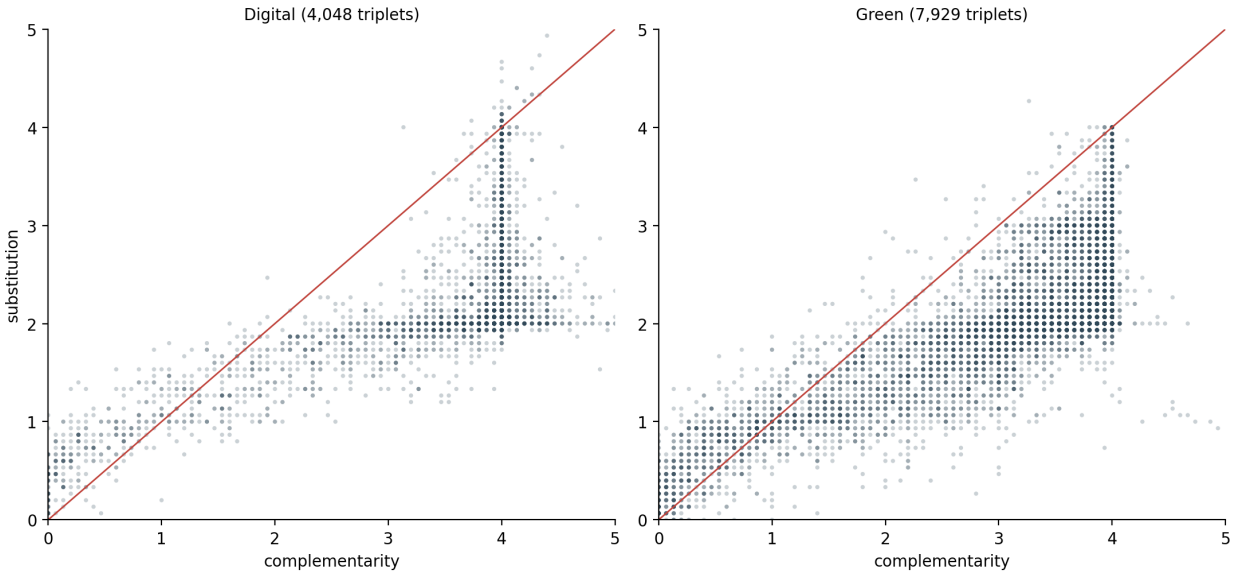


Figure 2: Triplet-level substitution against complementarity. Each point is a rated cell, and each coordinate is a mean over fifteen ratings. About 90 percent of the cells lie below the diagonal, and substitution rises with complementarity. Substitution exceeds complementarity in 354 of 4,048 digital and 795 of 7,929 green triplets.

3.4 From ratings to substitution and complementarity content

The ratings become a measure by reading each answer as a fraction of the task content the technology reaches and multiplying that fraction by the exposure. The three agreeing grades count as half, three quarters and all of the reached content, following their scale labels. The disagreeing grades and the zero count as none, so that a disagreeing answer never adds content, although the label of the higher disagreeing grade names a quarter. Section 5.5 shows the zero and the lowest grade to be the same answer. Averaged over the fifteen ratings of a triplet the mapping gives a substitution fraction $\sigma_{n,o,i}$ and a complementarity fraction $\kappa_{n,o,i}$ for technology n, occupation group o and section i. Across the triplets σ is zero in about half and averages below a fifth, and κ averages above one half and is zero in a tenth or fewer. The mapping lowers the Spearman rank correlation between the two constructs from 0.62 and 0.77 on the raw scale to about one half between the two fractions, and σ exceeds κ in about one triplet in a hundred.

An entity's content is its exposure to each technology times the rated fraction, summed over technologies. For a four-digit occupation e in group $o(e)$, with $x_{en}(w)$ its exposure to technology n over the window w of the application and N the number of technology nodes of the domain,

$$\Sigma_e \;=\; \frac{1}{N}\sum_{n}\sum_{i}\omega_{o(e)i}\;x_{en}(w)\,\sigma_{n,o(e),i}, \qquad \Gamma_e \;=\; \frac{1}{N}\sum_{n}\sum_{i}\omega_{o(e)i}\;x_{en}(w)\,\kappa_{n,o(e),i}, \tag{1}$$

where $\omega_{o(e)i}$ is the pooled employment share of section i among the group's workers in the LFS, and cells outside the grid contribute nothing, so that a technology whose exposure to the pair fell below the thresholds is counted as reaching no content worth rating. The industry-side content of a NACE

code is the mirror image. Its exposure to each technology is multiplied by the fraction rated for its section and averaged over the occupations employed in the section, weighted by their employment shares. Σ_e corresponds to the displacement of workers from tasks in the framework of Acemoglu and Restrepo (2019), and Γ_e to the augmentation of work that stays with workers, the distinction that Autor et al. (2024) draw for patents. The rating supplies the direction and the exposure the magnitude. Because the two constructs are rated separately, σ and κ are separate intensities. For robot manipulators both are high, and $\Sigma_e + \Gamma_e$ need not equal the exposure and can exceed it, up to twice its value. The contents are also computed at the level at which exposure is measured, so they aggregate along the same paths as the exposure they accompany, to sections and industry groups on the industry side and to occupation groups on the occupation side, and can take the place of exposure in specifications that use it.

Table 3 ranks technologies by the substitution share of their assessed effect, $\bar{\sigma}_n/(\bar{\sigma}_n + \bar{\kappa}_n)$ over the technology's rated triplets. The digital ranking is led by robotics and artificial-intelligence technologies, service robotics, robot manipulators, autonomous farm equipment, symbolic AI and agent systems and language models, and ends with infrastructure and interface technologies, messaging middleware, operating-system kernels, graphical user interfaces, immersive technologies and display panels. The green ranking runs from the production-process technologies of cement, metals, chemicals and glass to energy efficiency in buildings, water conservation, health protection and conventional vehicles. The appearance of cement and glass production near the top reflects the labeling problem of Section 5.5, in which these nodes are rated as industrial processes and are not read as mitigation technologies. A share of two small fractions is unstable. Table 3 therefore reports both fractions, and glass and cement production rank high on fractions below a quarter. Across technologies the share has a standard deviation of 0.13 in both domains, and within a technology, across the sections in which it was rated, about half that or less.

Substitution and complementarity content are correlated. Across the 427 occupations the correlation is 0.67 for digital and 0.89 for green technologies, and across the 272 NACE codes it is 0.68 and 0.94. Both contents also correlate with exposure, at 0.71 to 0.88. Because the two contents enter the employment regression of Section 4.3 jointly, the green coefficients rest on little independent variation. Digital substitution content is zero for 19 percent of occupations, mostly those with no rated cell.

4 Validation of the direction measure

4.1 Stability and determinants of direction

For each triplet the direction is the difference between the complementarity and the substitution fraction, $\kappa_{n,o,i} - \sigma_{n,o,i}$. It is positive in about nine triplets in ten. When each of the fifteen ratings is assigned at random to one of two halves, the directions computed from the two halves correlate above 0.9 in both domains, a split-half stability of about 0.95 by the Spearman–Brown formula (Table 4). At the low sampling temperature the draws vary little by design, so the statistic captures the consistency of the sampled ratings and says nothing about agreement between independent raters. The rating noise in a triplet's direction has a standard deviation of 0.05. Table 4 decomposes the variance of $\kappa - \sigma$ across triplets. The technology alone accounts for the largest share in the digital domain, the technology and the occupation for similar shares in the green domain, and the industry alone for 5 percent or less. The combination of technology and occupation accounts for about four fifths of the variance, 0.81 and 0.78. Pairs rated in a single industry are fitted exactly by the pair effects, but they hold only 316 of the digital and 448 of the green triplets, and on the pairs rated in two or more industries the share is the same, 0.81 and 0.77. Within those pairs the industry and

Table 3: Technologies ranked by the substitution share of their assessed effect. The share is $\bar{\sigma}_n/(\bar{\sigma}_n+\bar{\kappa}_n)$ over the technology's rated triplets, and the table lists the top and bottom five per domain among technologies with at least twenty rated triplets.

Technology	$\bar{\sigma}_n$	$\bar{\kappa}_n$	Share	Triplets
Digital, highest				
Service Robotics	0.65	0.74	0.47	70
Robot Manipulators and Arm Systems	0.59	0.72	0.45	82
Autonomous Farm Equipment	0.23	0.29	0.44	26
Symbolic AI, Knowledge and Agent Systems	0.51	0.74	0.41	83
NLP, Generative AI and Language Models	0.51	0.76	0.40	92
Digital, lowest				
Messaging Middleware	0.00	0.68	0.00	24
OS Kernel, Process and Memory Management	0.00	0.49	0.01	32
Graphical User Interfaces and UX Design	0.02	0.85	0.02	23
AR/VR and Immersive Technologies	0.03	0.72	0.04	55
Display Panels	0.03	0.70	0.04	37
Green, highest				
Production of cement	0.16	0.22	0.43	22
Technologies in the production process for final industrial or consumer products	0.41	0.58	0.42	322
Technologies related to metal processing	0.36	0.51	0.41	124
Technologies relating to chemical industry	0.37	0.54	0.41	68
Glass production	0.07	0.12	0.38	43
Green, lowest				
Conventional vehicles (based on internal combustion engine)	0.00	0.23	0.00	224
In human health protection, e.g. against extreme weather	0.00	0.53	0.00	214
Water conservation; Efficient water supply; Efficient water use	0.01	0.59	0.01	197
Energy efficiency in buildings	0.00	0.44	0.01	409
Fuel efficiency-improving vehicle design	0.00	0.34	0.01	38

the rating noise together account for 19 and 23 percent of the variance, so the industry in which a technology meets an occupation accounts for at most about a fifth. Among the roughly 1,000 digital and 1,600 green technology-occupation pairs rated in two or more industries, the direction varies across industries by a median range of 0.17 and 0.20, three to four times the rating noise, and the range exceeds 0.2 in 41 and 50 percent of the pairs, but the direction changes sign beyond two noise standard deviations in one pair in five hundred or fewer. The largest differences run from net substitution or no relation in one industry to strong complementarity in another, as for electronic payment processing among keyboard operators, negative in information and communication and strongly positive in accommodation and food service activities.

Table 4: Stability and determinants of the direction $\kappa - \sigma$ across rated triplets.

	Digital	Green
A. Ratings		
Triplets	4,048	7,929
Mean direction $\kappa - \sigma$	0.459	0.381
Share of triplets with $\kappa > \sigma$	0.90	0.89
Split-half correlation	0.931	0.908
Split-half reliability (Spearman-Brown)	0.964	0.952
Standard deviation of the rating noise in $\kappa - \sigma$	0.051	0.053
B. Share of the variance of $\kappa - \sigma$ explained by fixed effects		
Technology	0.430	0.273
Occupation	0.231	0.247
Industry	0.040	0.048
Technology, occupation and industry, additive	0.585	0.496
Technology × occupation	0.812	0.780
Technology × industry	0.538	0.400
Occupation × industry	0.311	0.315
C. Technology-occupation pairs rated in two or more industries		
Pairs	999	1,586
Triplets in pairs rated in one industry	316	448
Technology × occupation, R^2 on these pairs	0.810	0.774
Share of the variance of $\kappa - \sigma$ within pairs	0.190	0.226
Standard deviation of $\kappa - \sigma$ across industries (mean over pairs)	0.103	0.110
Median range of $\kappa - \sigma$ across industries	0.167	0.200
Pairs with a range above 0.2 (percent)	40.6	49.6
Pairs changing sign beyond two noise standard deviations (percent)	0.2	0.1

Direction is $\kappa - \sigma$ per technology-occupation-industry triplet, with each fraction the mean over fifteen ratings mapped as in Section 3.4. For the split halves, the ratings of each triplet and construct are divided at random into two halves, and the noise standard deviation is half the standard deviation of the difference between the halves' directions. Panel B reports the R^2 of $\kappa - \sigma$ on the fixed effects named. Panel C covers technology-occupation pairs rated in at least two sections, and a sign change requires $\kappa - \sigma$ above twice the noise standard deviation in one section and below its negative in another.

4.2 Aggregation and the direction signal

Table 5 reports the two contents at each level of aggregation. Across NUTS-2 regions (the EU's second-level statistical regions) their standard deviation is a quarter or less of that at any other level, and across the ten industry groups from which regional values are built they correlate at 0.88

for digital and 0.91 for green. Regional values vary little and are built from ten group values, so regions cannot separate the two contents. Across three-digit occupation groups they correlate at 0.61 for digital and 0.88 for green.

Table 5: The two contents by level of aggregation, with exposure over 2009–2023. Units without rated content are counted at zero. Three-digit occupation groups, NACE sections and the ten industry groups are unweighted means of their four-digit occupations or three-digit industries, and regional values weight the ten group values by each region's 2009 employment shares.

		Substitution content		Complementarity content		
Level	Units	Mean	Standard deviation	Mean	Standard deviation	Correlation of the two contents
A. Digital						
Four-digit occupations	427	0.012	0.016	0.041	0.060	0.67
Three-digit occupation groups	127	0.012	0.014	0.042	0.063	0.61
Three-digit NACE industries	272	0.010	0.012	0.034	0.062	0.68
NACE sections	21	0.006	0.008	0.028	0.051	0.80
Ten industry groups	10	0.007	0.008	0.040	0.070	0.88
NUTS-2 regions	313	0.007	0.001	0.027	0.004	0.66
B. Green						
Four-digit occupations	427	0.157	0.192	0.571	0.514	0.89
Three-digit occupation groups	127	0.147	0.190	0.545	0.503	0.88
Three-digit NACE industries	272	0.218	0.176	0.701	0.497	0.94
NACE sections	21	0.151	0.202	0.566	0.625	0.97
Ten industry groups	10	0.119	0.130	0.502	0.448	0.91
NUTS-2 regions	313	0.145	0.032	0.551	0.099	0.97

4.3 Substitution and complementarity content and employment change

For occupation group o in country c over a period from t_0 to t_1, Table 6 estimates

$$\ln E_{co,t_1} - \ln E_{co,t_0} \;=\; \beta_\Sigma \, \Sigma_o + \beta_\Gamma \, \Gamma_o + \phi \ln E_{co,t_0} + \mu_{\text{ISCO1}} + \delta_c + \varepsilon_{co}, \tag{2}$$

where $E_{co,t}$ is employment, corrected for the 2021 break in the survey series, Σ_o and Γ_o the substitution and complementarity contents of one domain, built from the patents filed in the years of the period, averaged over the four-digit codes of the group and divided by their standard deviation across cells, μ_{ISCO1} fixed effects for the ISCO major group and δ_c country fixed effects. The digital and the green pair are estimated in separate regressions, weighted by base-year employment, with standard errors clustered by occupation group, on the cells observed in all three periods. Under the displacement reading of the task framework $\beta_\Sigma < 0$, and under the augmentation reading $\beta_\Gamma > 0$. Occupation series at this detail begin in 2011, with ISCO-08, so the full period is 2011–2023, which the table also splits into 2011–2019 and 2020–2023. Columns (1) to (3) use the LFS microdata cells at three digits for 21 countries, the 24 countries of the microdata extract less Bulgaria, Slovenia and Malta, which report occupation at one or two digits. Germany, Austria and the Netherlands are absent from the extract. Columns (4) to (6) use the public Eurostat table of employment by two-digit occupation for the same 21 countries, so that the test can be reproduced without access to the microdata.

Over 2011–2023 in the microdata, one standard deviation more digital complementarity content is associated with employment growth about 20 log points higher and one standard deviation more

digital substitution content with growth about 12 log points lower, both significant at the 1 percent level. Green substitution content is negative and not significant (Table 6). Over 2011–2023 in the public table only the green substitution content is significant. Of the two sub-periods, the association appears in 2020–2023. Over 2020–2023 both digital contents carry their expected signs and are significant in both sources, with coefficients about half as large in the public table, and the green substitution content is negative in both and significant in the public table. Over 2011–2019 no substitution content is significant, and the complementarity contents are significant only at 10 percent, in one source each. Extending the public table to all member states lowers the digital estimates for 2020–2023 to −0.035 and 0.027, both still significant. Without Germany, whose occupation series carries a break flag in the Eurostat table, they are −0.043 and 0.035, close to those on the 21 countries. The two sub-periods omit the change from 2019 to 2020, so columns (1) and (2) do not add up to column (3). Exposure alone is not associated with employment change in any period or source, and when the domain exposure is entered next to the two contents, substitution keeps its sign and significance over 2020–2023 in both sources and complementarity in the microdata, while over 2011–2023 in the microdata the three regressors, correlated at 0.7 to 0.9, separate less cleanly (Section 5.7). Coding the second grade of the scale as a quarter of the reached content instead of none makes the 2020–2023 associations larger.

The respondent model was trained on text that postdates the outcome periods and may describe the occupational changes this test measures, so the test is not out of sample for the rater. If the ratings reflected knowledge of past employment trends, the association would appear in both periods, but over 2011–2019 no substitution content is significant and the complementarity contents only at 10 percent. This does not establish that the rater was uninformed by the outcomes.

Table 6: Substitution and complementarity contents and employment change across country-by-occupation cells. Employment change is over 2011–2019, 2020–2023 and 2011–2023, in the LFS microdata at three-digit ISCO in columns (1) to (3) and in the public Eurostat table at two-digit ISCO in columns (4) to (6), and the contents are built from the patents filed in the years of each period.

	Microdata, ISCO three-digit			Public table, ISCO two-digit		
Change in log employment	2011–2019	2020–2023	2011–2023	2011–2019	2020–2023	2011–2023
	(1)	(2)	(3)	(4)	(5)	(6)
A. Digital contents						
Digital substitution content	0.009	-0.081***	-0.125***	0.055	-0.044**	-0.023
	(0.027)	(0.017)	(0.046)	(0.051)	(0.021)	(0.059)
Digital complementarity content	0.055*	0.067***	0.196***	0.010	0.033**	0.064
	(0.033)	(0.013)	(0.048)	(0.036)	(0.015)	(0.043)
B. Green contents						
Green substitution content	0.022	-0.046	-0.088	-0.038	-0.047**	-0.088**
	(0.031)	(0.029)	(0.066)	(0.028)	(0.020)	(0.037)
Green complementarity content	0.003	0.007	0.067	0.061*	-0.001	0.068
	(0.039)	(0.036)	(0.068)	(0.034)	(0.029)	(0.045)
ISCO major-group fixed effects	✓	✓	✓	✓	✓	✓
Country fixed effects	✓	✓	✓	✓	✓	✓
Log base-year employment	✓	✓	✓	✓	✓	✓
Base-year employment weights	✓	✓	✓	✓	✓	✓
Occupation clusters	120	120	120	40	40	40
Observations	1,739	1,739	1,739	786	786	786

Change in log employment of country-by-occupation cells observed in all three periods, on the contents built from the patents filed in the years of the same period, averaged over the four-digit codes of the occupation group and divided by their standard deviation. The digital and the green pair are estimated in separate regressions. Columns (1) to (3) use the EU Labour Force Survey microdata and columns (4) to (6) the public Eurostat table of employment by two-digit occupation, for the same 21 EU countries. Occupation series begin in 2011, with ISCO-08. Standard errors clustered by occupation group are in parentheses. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

5 Design choices and further results of the direction measure

5.1 The rating prompt

Each call sends a system message, which sets the model's standing instructions, and a user message, which poses the question. The placeholder `{domain}` is digital or green, `{Domain}` its capitalized form at the start of a line, and the bracketed fields are filled from the triplet. The system message is reproduced below.

```
You are an expert in {domain} technologies. Your expertise is defined by the
given persona and context in user prompt.
Task: Given a prompt, produce exactly ONE JSON object evaluating the statement.

STRICT OUTPUT FORMAT:
- Output MUST be a single JSON object with exactly these keys in this order:
  1) "rating": an integer from 0 to 5 (no decimals).
  2) "reason": a 1-3 lines string (<= 25 words), plain text.
- No additional keys, no trailing commas.
- No prose before or after the JSON. No code fences. No comments. No markdown.

POLICY:
- If information is ambiguous, choose the closest rating and explain briefly.
- Never ask questions or request more info.
- Use a neutral, expert tone; avoid hedging unless required by the persona.
- 0 means you do not know. Never use 0 to signal a refusal.
```

The user message is reproduced below.

```
You are assessing how a {domain} technology will affect an occupation as it is
practised within a specific industry.
- Substitute: the {domain} technology takes over the work itself, so fewer
  human workers are needed to do it.
- Complement: the {domain} technology supports or enhances the human workers,
  who still do the work, but better, faster or at higher quality.

You are {role}. Answer strictly from that perspective.

Occupation: [occupation group title]
Industry: [NACE section title]
{Domain} technology: [technology name or ENV-TECH description]

To what extent do you agree with the statement below?
- 1 = Strongly disagree
- 2 = Disagree (25 percent)
- 3 = Neutral / somewhat agree (50 percent)
- 4 = Mostly agree (75 percent)
- 5 = Fully agree
- 0 = Do not know

Statement: {statement}

Provide exactly one rating from [0, 1, 2, 3, 4, 5] and one concise reason
(1-3 sentences).
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The role is "a manager who runs the team that performs this work day to day", "an experienced worker who personally performs this task day to day" or "a chief executive officer responsible for the

organization's strategy and investment decisions". The substitution statement is "This {domain} technology will replace human workers in this occupation within this industry." and the complementarity statement "This {domain} technology will work alongside human workers in this occupation within this industry and support them." The response is constrained by a JSON schema with an integer rating from 0 to 5 and a string reason.

5.2 The prompt-format experiment

Whether consolidated prompt formats lose information was tested on twelve triplets drawn from the exposure grid, one central and one peripheral triplet for each of the three highest-exposure technologies of each domain. Each of four formats with identical wording was run with five draws, for 720 calls and 1,440 ratings in total. The four formats are the separated format, with one construct and one persona per call, a format merging the two constructs, a format merging the three personas, and a fully merged format. The experiment was run before the production prompt was fixed, at temperature and top-p of 1.0, with a horizon of 2030, a human-resources manager in place of the chief executive and the zero defined as do not know or no effect, on triplets from an earlier exposure build. The comparison holds the prompt fixed across formats. Table 7 reports it. Agreement with the separated reference declines monotonically with consolidation, and individual cell means shift by up to three scale points. Use of the no-effect category falls when personas share a call, from 7.8 to 0.3 percent overall with merged personas, and on the single no-effect triplet in the test set from 36.7 to 3.3 percent. The latter comparison rests on one triplet and shows the sign of the change without estimating its size. Draw-to-draw variability increases monotonically with consolidation, and the merged-construct format widens the substitution-complementarity gap by half a scale point, which would bias the measured direction directly. The separated format was therefore retained despite its sixfold cost.

Table 7: Prompt-format experiment on twelve triplets, with five draws per format. The separated format is the reference.

	Separated	Merged constructs	Merged personas	Fully merged
Calls per triplet-draw	6	3	2	1
Correlation with separated cell means	1	0.883	0.805	0.764
Mean absolute deviation		0.42	0.52	0.53
Zero share, all ratings (%)	7.8	4.2	0.3	3.1
Zero share, no-effect triplet (%)	36.7	33.3	3.3	10.0
Within-prompt draw SD	0.475	0.523	0.587	0.646
Net rating (substitution minus complementarity), mean	−1.056	−1.572	−0.811	−1.146

5.3 The grid design space

The grid is one point in a design space with six dimensions. These are occupation resolution (three- or four-digit ISCO), industry resolution (NACE section, division or group), technology resolution (base nodes, intermediate levels, or one indicator per domain), the exposure filter (none, top-K per margin, or percentile cuts), the source recording occupation-industry pairs (the LFS, the OECD Survey of Adult Skills, online vacancies, or none), and the co-occurrence floor. The adopted design combines the finest resolution at which a representative EU-wide employment source records pairs, three-digit occupations by sections under the LFS, with the common exposure thresholds

of Section 3.2, which are 1.31 for digital occupations, 1.78 for digital industries, 3.79 for green occupations and 4.15 for green industries. The 0.1 percent employment floor is the co-occurrence cut in the LFS, where presence alone admits pairs with negligible employment. Four-digit occupations could be recorded only through the Survey of Adult Skills (PIAAC), which covers thirteen EU countries and in which any share-based floor reduces to requiring that the pair be observed at all, because the median occupation counts about 140 respondents.

5.4 Personas

The personas move the level of the ratings and leave their ranking. The chief executive returns the highest mean substitution rating in both domains, 2.38 in the digital and 2.06 in the green, and the worker the lowest, 2.10 and 1.85, while complementarity differs little across roles, from 3.50 to 3.43 in the digital and from 3.07 to 2.88 in the green domain (Figure 3). Across triplets the three persona means correlate between 0.88 and 0.95 and differ in absolute value by 0.17 to 0.35 scale points. In a decomposition of the variance of the individual ratings, between 70 and 84 percent lies between triplets, 4 to 7 percent between the personas of one triplet, and 12 to 23 percent between draws of one persona.

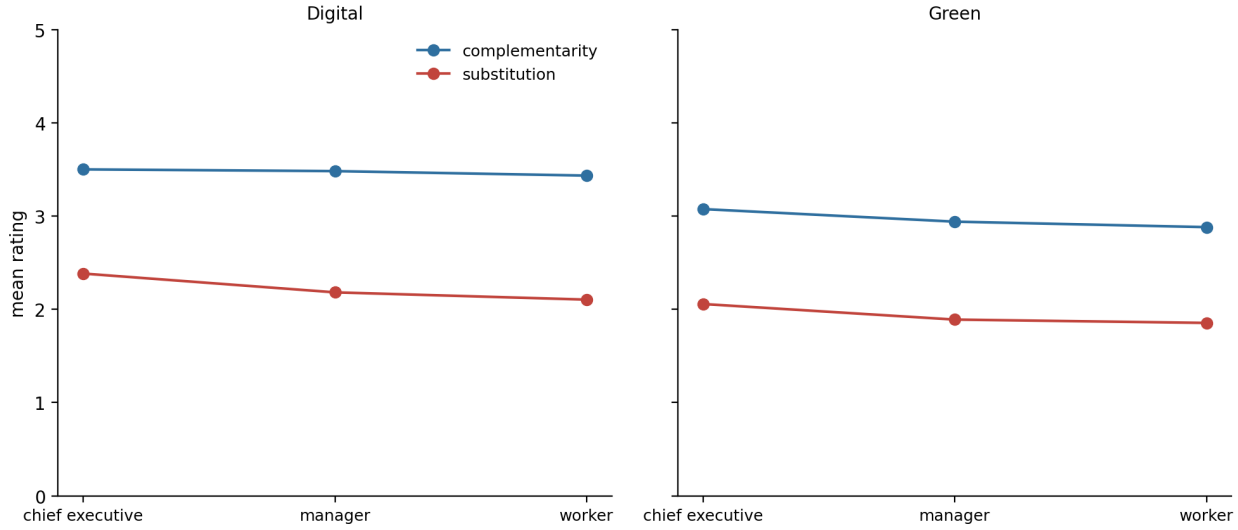


Figure 3: Mean rating by persona, construct and domain. The three roles differ by at most 0.28 scale points on substitution and 0.19 on complementarity, with the chief executive highest on substitution and the worker lowest in both domains.

5.5 How the zero category is used

Zeros are 9.4 percent of all ratings, 7.4 and 3.4 percent for digital complementarity and substitution and 13.4 and 9.6 percent for their green counterparts, and the worker uses the category almost twice as often as the chief executive, in 12.1 against 6.9 percent of answers. Although the scale defines the zero as do not know, the justifications, which we read for every zero rating, show that it functions as a no-effect code. Outside the twelve green nodes discussed below, which account for 30.8 percent of the zeros, the stated reason is typically that the technology has no relation to the work of that occupation in that industry, which is the content of a rating of 1. The two categories are interchangeable at the level of the individual response. Section 5.7 reports the employment

test with the second grade, whose label names a quarter, coded as a quarter instead of none. The same persona rating the same triplet returns the same justification as a zero in one draw and as a 1 in another. For additive manufacturing with financial and mathematical associate professionals in accommodation and food service, the worker's zero reads "I see no direct connection between 3D printing and my daily tasks like budgeting or data analysis" and the worker's 1 "I see no direct link between 3D printing and my core tasks like budgeting or revenue analysis. It does not support or enhance my daily accounting or reporting work." A zero is therefore retained as a zero, and the substitution and complementarity shares, S and C, are means over all fifteen ratings per construct on a scale running from 0 to 5, with the zero as the bottom category.

The exception is a set of twelve green nodes whose ENV-TECH descriptions are not readable as technologies once detached from their parent heading. Six name an activity that is not itself green, among them conventional vehicles, aeronautics or air transport, glass production and nuclear fission reactors, and six are umbrella descriptions such as enabling technologies with a potential contribution to emissions mitigation or technologies having an indirect contribution to adaptation. For these nodes the zero records that the label cannot be rated and does not mean that the technology leaves the work unaffected. Their zeros are 1.7 and 1.2 percent of all ratings respectively, 2.9 percent together, while the twelve nodes carry 19 percent of all ratings. Section 5.6 lists them with their zero shares and the justifications on which the reading rests.

5.6 Green nodes not readable as technologies

The justifications of the zero ratings separate the twelve nodes into two groups. In the first, the ENV-TECH description names an activity that is not green once separated from its parent heading on climate change mitigation, and the respondent rejects the premise of the question. For glass production rated for refuse workers in water supply and waste management, one justification reads "Glass production is not a green technology used in waste management; it is a separate manufacturing process." In the second, the description is an umbrella formulation that names no technology. For technologies with an indirect contribution to adaptation rated for drivers in manufacturing, one justification reads "The technology is too vaguely defined to assess its direct impact on driving tasks in manufacturing." Table 8 reports the twelve nodes. The nodes are retained with the ratings as given, and the reading of the zero adopted in Section 3.3 applies to the zeros outside these nodes, about 69 percent of all zeros.

5.7 Exposure, direction and the coding of the scale

Table 9 re-estimates the employment test of Table 6 in three variants. The first enters the domain exposure of the occupation group alone. The second enters it next to the two contents. The third keeps the two contents and codes the second grade of the scale, whose label names a quarter, as a quarter of the reached content instead of none. Exposure alone is not associated with employment change in any period or source. Entered next to the contents, with which it correlates at 0.7 to 0.9, it leaves substitution significant over 2020–2023 in both sources and complementarity in the microdata, while over 2011–2023 in the microdata the three regressors separate less cleanly, with exposure negative and substitution no longer significant. The alternative coding of the second grade makes the contents larger, strengthens the digital and the green associations over 2011–2023 and 2020–2023, and leaves 2011–2019 without a significant substitution content.

Table 8: The twelve green nodes whose descriptions are not readable as technologies, with their use of the zero category. Together they carry 30.8 percent of all zero ratings, and their zeros are 2.9 percent of all ratings.

Node description	Triplets	Ratings	Zeros	Share (%)
Activity not readable as green without its parent heading				
Conventional vehicles (based on internal combustion engine)	224	6,720	1,775	26.4
Aeronautics or air transport	107	3,210	1,491	46.4
Glass production	43	1,290	883	68.4
Production of cement	22	660	212	32.1
Nuclear fission reactors	85	2,550	1,448	56.8
Ceramic production	23	690	363	52.6
Umbrella description naming no technology				
Technologies having an indirect contribution to adaptation to climate change	379	11,370	1,723	15.2
Enabling technologies with a potential contribution to GHG emissions mitigation	812	24,360	730	3.0
At coastal zones; at river basins	60	1,800	930	51.7
Enabling technologies or technologies with a potential or indirect contribution to GHG emissions mitigation	132	3,960	193	4.9
Technologies in the production process for final industrial or consumer products	322	9,660	587	6.1
Climate change mitigation technologies for sector-wide applications	88	2,640	91	3.4

Table 9: Employment change on exposure alone, on exposure with the two contents, and on the contents with the second grade coded as a quarter.

	Microdata, ISCO three-digit			Public table, ISCO two-digit		
Change in log employment	2011–2019	2020–2023	2011–2023	2011–2019	2020–2023	2011–2023
	(1)	(2)	(3)	(4)	(5)	(6)
A. Digital						
Digital exposure alone	0.022	0.006	0.012	0.040	0.005	0.036
	(0.030)	(0.020)	(0.059)	(0.030)	(0.020)	(0.041)
Digital exposure, with the contents	-0.273**	-0.077	-0.536**	-0.072	0.026	-0.038
	(0.137)	(0.063)	(0.220)	(0.142)	(0.052)	(0.168)
Substitution content, with exposure	0.046	-0.067***	-0.032	0.069	-0.051**	-0.014
	(0.034)	(0.021)	(0.066)	(0.066)	(0.022)	(0.075)
Complementarity content, with exposure	0.289**	0.133**	0.658***	0.070	0.012	0.094
	(0.123)	(0.053)	(0.195)	(0.115)	(0.044)	(0.138)
Substitution content, second grade as a quarter	0.023	-0.180***	-0.324***	0.140	-0.089*	-0.040
	(0.068)	(0.037)	(0.114)	(0.128)	(0.049)	(0.146)
Complementarity content, second grade as a quarter	0.038	0.185***	0.424***	-0.087	0.088**	0.087
	(0.072)	(0.033)	(0.111)	(0.117)	(0.041)	(0.132)
B. Green						
Green exposure alone	0.010	-0.037	-0.037	0.046	-0.038	0.003
	(0.023)	(0.025)	(0.047)	(0.034)	(0.028)	(0.041)
Green exposure, with the contents	-0.014	-0.058	-0.137	0.057	-0.047	-0.030
	(0.053)	(0.037)	(0.094)	(0.073)	(0.032)	(0.061)
Substitution content, with exposure	0.019	-0.058**	-0.119*	-0.021	-0.060***	-0.096**
	(0.034)	(0.029)	(0.071)	(0.023)	(0.021)	(0.037)
Complementarity content, with exposure	0.020	0.071	0.218	-0.001	0.049	0.100
	(0.077)	(0.052)	(0.133)	(0.073)	(0.038)	(0.068)
Substitution content, second grade as a quarter	0.011	-0.125**	-0.299**	-0.081	-0.101**	-0.194**
	(0.068)	(0.058)	(0.130)	(0.061)	(0.044)	(0.085)
Complementarity content, second grade as a quarter	0.013	0.086	0.272**	0.108*	0.053	0.177**
	(0.072)	(0.062)	(0.127)	(0.062)	(0.047)	(0.085)
Observations	1,739	1,739	1,739	786	786	786

Change in log employment of country-by-occupation cells in the design of Table 6, with the same cells, fixed effects, control, weights and standard errors. The domain exposure is the occupation group's mean exposure over the domain's technologies, built from the patents of the same period and divided by its standard deviation across cells. The first two rows of each panel enter exposure alone and then next to the two contents, whose coefficients follow. The last two rows re-estimate the contents of Table 6 with the second grade of the scale coded as a quarter of the reached content instead of none. * p<0.10, ** p<0.05, *** p<0.01.

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